HERSCHEL ISLAND, YT, Canada

WMO: **715010**

Lat: **69.568N** Lon: **138.911W** Elev: **1** StdP: **101.31** Time Zone: **-8.00 (NAP)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB 99.6%	Heating DB 99%	Humidification 99.6% DP	99.6% HR	99.6% MCDB	99% DP	99% HR	99% MCDB	Coldest Month 0.4% WS	0.4% MCDB	1% WS	1% MCDB	MCWS/PCWD to 99.6% DB MCWS	PCWD	WSF	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	-35.8	-34.2	-39.6	0.1	-35.8	-37.8	0.1	-34.3	23.1	-18.1	20.3	-16.9	3.2	320	1.034	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling 0.4% DB	0.4% MCWB	1% DB	1% MCWB	2% DB	2% MCWB	Evaporation 0.4% WB	0.4% MCDB	1% WB	1% MCDB	2% WB	2% MCDB	MCWS/PCWD to 0.4% DB MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	5.8	17.0	14.0	14.9	12.6	13.2	11.5	14.5	16.7	12.9	14.6	11.6	13.1	4.0	110	(2)

	Dehumidification 0.4% DP	0.4% HR	0.4% MCDB	1% DP	1% HR	1% MCDB	2% DP	2% HR	2% MCDB	Enthalpy 0.4% Enth	0.4% MCDB	1% Enth	1% MCDB	2% Enth	2% MCDB	Extreme Max WB	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	13.4	9.6	15.4	11.8	8.6	13.9	10.5	7.9	12.8	40.7	16.7	36.4	14.7	33.1	13.2	25.3	(3)

Extreme Annual Design Conditions

	Extreme Annual WS 1%	2.5%	5%		Extreme Annual Temperature Mean Min	Mean Max	Standard Deviation Min	Standard Deviation Max	n=5 years Min	n=5 years Max	n=10 years Min	n=10 years Max	n=20 years Min	n=20 years Max	n=50 years Min	n=50 years Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	16.0	13.4	11.7	DB	-37.4	21.3	2.1	3.0	-39.0	23.4	-40.2	25.1	-41.4	26.8	-42.9	29.0	(4)
(5)				WB	-37.9	15.8	2.0	1.5	-39.4	16.9	-40.6	17.7	-41.7	18.6	-43.2	19.7	(5)

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.6	-23.2	-23.6	-21.5	-13.2	-3.2	4.1	9.2	6.8	2.1	-5.4	-15.0	-21.0	(6)
(7)		DBStd	13.00	7.00	7.08	6.59	6.26	4.50	3.35	3.74	3.71	2.95	4.72	5.96	6.05	(7)
(8)		HDD10.0	6832	1028	942	976	697	411	181	61	113	236	478	749	960	(8)
(9)		HDD18.3	9821	1286	1175	1234	947	669	428	284	357	486	737	999	1218	(9)
(10)		CDD10.0	53	0	0	0	0	0	3	36	14	0	0	0	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH23.3	5	0	0	0	0	0	0	5	0	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.0	5.7	5.1	5.0	4.3	4.4	4.5	4.8	4.7	5.1	5.7	5.5	5.1	(14)
(15)	Precipitation	PrecAvg	273	11	8	10	7	13	33	50	55	34	24	15	12	(15)
(16)		PrecMax	332	30	19	29	24	29	63	108	103	62	41	36	27	(16)
(17)		PrecMin	203	1	1	1	1	2	9	20	20	17	10	2	3	(17)
(18)		PrecStd	31	7	4	6	5	6	16	17	18	10	8	7	6	(18)

				Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.6	-3.2	-0.8	2.9	7.3	16.4	23.3	18.7	11.5	5.0	0.8	-3.5	(19)
(20)			MCWB	-2.6	-3.4	-1.8	1.6	5.1	12.5	18.3	14.3	8.5	3.4	-0.2	-4.1	(20)
(21)		2%	DB	-5.6	-7.1	-5.6	0.2	4.8	13.3	18.7	15.7	8.9	2.4	-2.3	-7.7	(21)
(22)			MCWB	-6.2	-7.6	-6.1	-0.2	3.2	10.9	15.6	12.9	7.3	1.6	-3.0	-8.3	(22)
(23)		5%	DB	-9.8	-10.0	-8.9	-1.9	3.3	11.0	16.1	13.7	7.2	1.1	-4.4	-10.1	(23)
(24)			MCWB	-10.2	-10.3	-9.2	-2.4	2.2	9.1	13.6	11.8	6.3	0.4	-4.9	-10.4	(24)
(25)		10%	DB	-13.3	-13.2	-11.7	-4.0	2.0	9.0	14.3	11.9	6.0	-0.2	-6.5	-12.6	(25)
(26)			MCWB	-13.6	-13.6	-12.0	-4.4	1.2	7.5	12.4	10.6	5.2	-0.8	-7.0	-13.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.0	-3.4	-1.9	1.7	5.7	13.4	20.4	15.0	9.0	3.7	-0.8	-4.1	(27)
(28)			MCDB	-1.5	-3.1	-1.0	2.8	6.7	15.2	21.9	17.7	11.0	4.8	0.1	-3.6	(28)
(29)		2%	WB	-6.0	-7.5	-5.8	-0.1	3.3	11.0	15.9	13.3	7.5	1.8	-3.2	-8.6	(29)
(30)			MCDB	-5.2	-7.1	-5.3	0.4	4.6	13.3	18.0	15.1	8.6	2.4	-2.6	-8.3	(30)
(31)		5%	WB	-9.8	-10.4	-9.1	-2.5	2.2	9.2	14.0	11.8	6.4	0.5	-5.1	-10.8	(31)
(32)			MCDB	-9.4	-10.0	-8.7	-2.2	3.2	10.8	16.0	13.4	7.2	1.0	-4.7	-10.5	(32)
(33)		10%	WB	-13.3	-13.5	-11.8	-4.6	1.2	7.6	12.6	10.6	5.2	-0.8	-7.3	-13.3	(33)
(34)			MCDB	-13.1	-13.2	-11.4	-4.3	1.9	9.0	14.2	11.9	5.9	-0.3	-6.8	-13.0	(34)
(35)	Mean Daily Temperature Range		MDBR	5.7	5.9	5.9	6.2	4.2	5.3	5.8	4.8	3.5	3.7	5.2	5.2	(35)
(36)		5% DB	MCDBR	9.8	10.3	8.6	7.6	6.7	9.1	9.0	8.3	6.2	4.4	6.4	8.0	(36)
(37)			MCWBR	9.5	10.4	8.1	7.0	5.3	6.8	6.1	5.4	4.7	3.5	6.0	8.3	(37)
(38)		5% WB	MCDBR	9.8	10.3	8.4	7.3	6.4	9.0	8.9	7.8	6.0	4.1	6.4	8.2	(38)
(39)			MCWBR	9.6	10.1	8.1	6.8	5.1	6.8	6.3	5.3	4.7	3.5	6.0	8.1	(39)

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
(40)	Clear-Sky Solar Irradiance	taub		0.167	0.207	0.242	0.303	0.339	0.349	0.370	0.340	0.301	0.226	N/A	N/A	(40)
(41)		taud		1.886	2.061	2.193	2.082	2.100	2.311	2.386	2.499	2.514	2.197	N/A	N/A	(41)
(42)		Ebn at Noon		32	559	792	822	833	842	807	798	735	526	0	0	(42)
(43)		Edh at Noon		3	40	72	112	126	106	96	77	56	34	0	0	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.01	0.39	2.01	4.30	5.92	6.15	4.87	3.11	1.65	0.49	0.00	0.00	(44)
(45)		RadStd		0.00	0.04	0.10	0.24	0.44	0.48	0.40	0.27	0.16	0.08	0.00	0.00	(45)

Historical Trends

		DBAvg	Heating 99% DB	Heating 99% DP	Cooling 1% DB	Cooling 1% WB	Cooling 1% DP	Degree-Days HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	+0.75	+0.94	N/A	N/A	N/A	N/A	-254	-274	N/A	N/A	(47)

Nomenclature: See separate page